

Finding Angle Measures in Parallelograms

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A **parallelogram** is a quadrilateral with two pairs of parallel sides.

Opposite angles of a parallelogram are congruent.

Consecutive angles of a parallelogram are supplementary.

Their measures add up to 180° .

Solving Angle Problems

Identify whether the angles are opposite or consecutive.

Write an equation using the appropriate property.

Solve for the variable.

Substitute the value back to find the angle measures.

Examples

Example 1

In parallelogram $WXYZ$, $\angle W = 112^\circ$.
Find $\angle X$, $\angle Y$, and $\angle Z$.

$$\angle X = 68^\circ$$

$$\angle Y = 112^\circ, \quad \angle Z = 68^\circ$$

Example 2

In parallelogram $PQRS$,

$$\angle P = (2x + 15)^\circ$$

$$\angle Q = (4x - 3)^\circ.$$

Since consecutive angles are supplementary,

$$(2x + 15) + (4x - 3) = 180.$$

Find x , $\angle P$, and $\angle Q$.

Example 3

In parallelogram $EFGH$,

$$\angle E = (5x + 8)^\circ$$

$$\angle G = (8x - 40)^\circ.$$

Opposite angles are congruent.

$$5x + 8 = 8x - 40.$$

Find x and $\angle E$.

Example 4

In parallelogram $QRST$, the ratio of consecutive angles is $4 : 5$.

Let the angles be $4x$ and $5x$.

$$4x + 5x = 180^\circ$$

Find all four angle measures.

Practice Exercises

Practice 1

In parallelogram $WXYZ$,

$$\angle W = 112^\circ.$$

Find $\angle X$, $\angle Y$, and $\angle Z$.

Practice 2

In parallelogram $PQRS$,

$$\angle P = (2x + 15)^\circ$$

$$\angle Q = (4x - 3)^\circ.$$

Find x , $\angle P$, and $\angle Q$.

Practice 3

In parallelogram $ABCD$,

$$\angle A = (7x - 5)^\circ$$

$$\angle B = (4x + 20)^\circ.$$

Find x .

Practice 4

In parallelogram $EFGH$,

$$\angle E = (5x + 8)^\circ$$

$$\angle G = (8x - 40)^\circ.$$

Find x and $\angle E$.

Practice 5

In parallelogram $QRST$, the ratio of consecutive angles is $4 : 5$.

Find the measures of all four angles.

Practice 6

In parallelogram $WXYZ$,

$$\angle W = \left(\frac{3x}{4} + 12 \right)^\circ$$

$$\angle X = \left(\frac{x}{2} + 48 \right)^\circ.$$

Find x , $\angle W$, and $\angle X$.

Activities

Activity 1

In parallelogram $ABCD$,

$$\angle A = 74^\circ.$$

Find $m\angle B$, $m\angle C$, and $m\angle D$.

Activity 2

In parallelogram $JKLM$,

$$\angle J = (4x + 6)^\circ$$

$$\angle L = (6x - 18)^\circ.$$

Find x and the measure of each angle.

Activity 3

In parallelogram $ABCD$,

$$\angle A = (8x - 14)^\circ$$

$$\angle B = (5x + 22)^\circ.$$

Find x .

Activity 4

In parallelogram $MNOP$, the ratio of $\angle M$ to $\angle N$ is $3 : 5$.

Find the measures of all four angles.

Activity 5

In parallelogram $QRST$, the ratio of consecutive angles is $7 : 11$.

Find the measures of all four angles.

Activity 6

In parallelogram $ABCD$,

$$\angle A = \left(\frac{x}{3} + 28 \right)^\circ$$

$$\angle C = \left(\frac{5x}{3} - 20 \right)^\circ.$$

Find x .

Thank you!